# The Strouhal Prize

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## 1 Introduction

Flow around a cylinder is a well established problem of interest in fluid dynamics [3, 4]. This has been examined extensively both experimentally [8, Figure 14.15] and computationally (see references in [5, 6]). It is widely believed that a Hopf bifurcation occurs near Reynolds number 50 [2, 7], where steady flow evolves into the Karmán vortex street [1, 11].

The temporal periodicity of the vortex street was first observed (see [1]) experimentally by Strouhal in 1878 [10], although footnote 1 in [1] suggests that a complete history may trace back to Leonardo da Vinci, as is the case with many areas of fluid mechanics.

The papers [9, 12] suggest that the vortex shedding in the Karmán vortex street is periodic in time, beginning around Reynolds number 50 (coinciding with the believed Hopf bifurcation [7, 2]) and continuing this periodicity up to Reynolds number 1000, but that this periodicity noticeably begins to break down above that.

The paper [9] also demonstrates that there are easily computable, steady solutions of the Navier–Stokes equations for Reynolds numbers above 50. These are also trivially periodic in time, since they are constant in time. But we are interested in non-constant-in-time, periodic solutions.

We know of no mathematical proof of the existence of a non-constant, periodic-in-time solution at the Reynolds numbers indicated, and we offer a prize of $1,000 for the publication of such a proof, or for a proof that no such periodic solutions exist.

### 1.1 Time-dependent Navier–Stokes

For incompressible fluid flow, the Navier–Stokes equations are as follows,

$$\mathbf{u}_t - \nu\Delta\mathbf{u} + \mathbf{u}\cdot\nabla\mathbf{u} + \nabla p = 0, \qquad \nabla\cdot\,\mathbf{u} = 0, \quad \text{in } \Omega, \tag{1}$$

where $\nu$ is the kinematic viscosity.

We set the boundary condition $\mathbf{u} = \mathbf{g}$ on $\partial\Omega$.

## 1.2 Domain and boundary conditions

The domain $\Omega$ is defined by

$$\Omega = \left\{(x, y) \ : \ |x| < \infty, \ |y| < \infty, \ x^2 + y^2 > 1\right\}. \tag{2}$$

This domain surrounds a cylinder with radius 1, centered at $(0, 0)$. The boundary of this cylinder is $\Gamma$, defined by

$$\Gamma = \left\{(x, y) \ : \ x^2 + y^2 = 1\right\}.$$

We take $\mathbf{g} = (1, 0)^t$ for $x^2 + y^2 >> 1$ and $\mathbf{g} = \mathbf{0}$ on $\Gamma$. For the infinite domain $\Omega$, the boundary condition on $\partial\Omega\backslash\Gamma$ is interpreted to mean $\mathbf{u} \to \mathbf{g}$ as $x^2 + y^2 \to \infty$. It could be that other boundary conditions at $\infty$ are appropriate.

Since the diameter of the cylinder is 2, and the difference in flow rate on the cylinder versus at infinity is 1 in magnitude, the standard definition for the Reynolds number $R$ yields $R = 2/\nu$.

## 1.3 Solution regularity

We are seeking a solution that conforms closely to what is observed in common experience, formal experiments, and computational simulations, namely a smooth classical solution. Since the boundary is smooth, there is no reason to expect any singularities. For example, it is sufficient if it is proved that $\mathbf{u}$ in $C^2$ in space and $p$ in $C^1$ in space, with $C^1$ regularity in time. We expect $\mathbf{u}$ and $p$ to be $C^\infty$ in both space and time, and any deviations from this would be interesting to report.

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